

Cyber Security Internship – Task 1: Lab Setup Report

Student Details

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Internship: Cyber Security Internship

Task: Task 1 – Lab Environment Setup

1. Objective

The objective of this task was to create a cybersecurity lab environment using virtual machines for learning and practicing cybersecurity concepts in a safe and isolated environment.

The lab environment consists of:

- Kali Linux (Attacker Machine)
 - Metasploitable2 (Target Machine)
 - Wireshark (Network Traffic Analysis Tool)
 - Oracle VirtualBox (Virtualization Platform)
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2. Tools Used

Tool	Purpose
Oracle VirtualBox	Creating and managing virtual machines
Kali Linux	Security testing workstation
Metasploitable2	Vulnerable target machine
Wireshark	Network packet capture and analysis

3. Kali Linux Setup

Kali Linux was installed and configured in Oracle VirtualBox.

Network configuration was verified using:

```
ip a
```

The system received:

- NAT Interface (Internet Access)
- Host-Only Interface (Lab Communication)

The Host-Only Adapter allows Kali Linux to communicate with Metasploitable2 within the isolated lab environment.

Screenshot

Insert:

kali-linux.png

4. Metasploitable2 Setup

Metasploitable2 was imported into Oracle VirtualBox and configured as the vulnerable target machine.

Network configuration was verified using:

```
ifconfig
```

The system successfully obtained an IP address from the Host-Only Network.

This machine will be used as the target system for future security testing activities.

Screenshot

Insert:

metasploitable2.png

5. Network Communication Verification

Communication between Kali Linux and Metasploitable2 was verified through the Host-Only Network.

The successful communication confirms:

- Correct VirtualBox configuration
 - Proper Host-Only networking
 - Successful connectivity between both virtual machines
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6. Wireshark Test Capture

Wireshark was used to capture network traffic generated between Kali Linux and Metasploitable2.

The capture showed:

- ICMP Echo Requests
- ICMP Echo Replies
- ARP Traffic
- Network Discovery Traffic

This confirms that Wireshark is functioning correctly and is capable of monitoring communication within the lab environment.

Screenshot

Insert:

wireshark-capture.png

7. Results

The cybersecurity lab environment was successfully created and tested.

Completed Tasks

- Kali Linux installed successfully
 - Metasploitable2 installed successfully
 - VirtualBox configured successfully
 - Host-Only Network configured successfully
 - Communication verified between virtual machines
 - Wireshark packet capture completed successfully
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8. Conclusion

A fully functional cybersecurity lab environment was successfully established using Oracle VirtualBox, Kali Linux, Metasploitable2, and Wireshark.

The lab environment is now ready for future cybersecurity activities such as network scanning, enumeration, vulnerability assessment, and security testing tasks.

Screenshots Included

1. kali-linux.png
2. metasploitable2.png
3. wireshark-capture.png